



Tomato disorders

Quick facts

- Tomato plants can develop disorders that distort plants and blemish fruits.
- Tomato disorders are generally caused by varietal, environmental, or management issues
- For most disorders, good nutrient management and watering practices will reduce occurrence of issues.
- Different tomato varieties may be more or less likely to develop certain disorders.
- You may have to try out different tomatoes before finding varieties that do well in your location. Seed catalogs often give information that can help you choose varieties that avoid problems.

Tomatoes are susceptible to many issues, some of which are [caused by pathogens](#), some of which are caused by mismatches between tomato growth and the environment. Below are some of the most common issues caused by plant nutrition and the Minnesota environment.

Blossom-end rot

Blossom-end rot is one of the most common tomato disorders seen in Minnesota. It affects tomato fruit, especially the first set of fruit.



How to tell blossom end rot from other tomato issues

- Affected fruit have a tan to black flattened spot at the blossom end of the fruit.

- Secondary fungi and bacteria can enter the blossom end rot area, resulting in further decay of the fruit.
- Blossom end rot can appear on fruit in any stage of development, but it is most common when fruit are one-third to one-half grown.
- The first fruit produced by the plant are often most severely affected.
- Larger slicer-type tomatoes are usually more prone to blossom end rot than cherry tomatoes.
- Fruit that develop later in the season on the same plant can be unaffected.



Biology

Blossom-end rot is caused by a calcium deficiency in the tomato plant. Although blossom end rot means that the plant does not have enough calcium within the developing fruit, it does not mean that there is a lack of calcium in the soil.

Often blossom end rot occurs as a result of several cultural or environmental factors that affect the plant's ability to take up calcium, such as:

- Fluctuations in soil moisture.
- Heavy applications of nitrogen fertilizer.
- Injury to roots.
- Hot weather and environmental conditions that cause plants to grow too rapidly.

The amount of calcium available to the plant decreases rapidly when there is too much potassium, magnesium, ammonium and sodium in the soil. These are all salts, and so they use similar pathways to enter plants. Extreme fluctuation in moisture can also reduce the availability of calcium salts needed by the plant.

Heavy applications of nitrogen fertilizers and abundant rain cause rapid plant growth and can cause blossom-end rot, especially during periods of dry, hot weather.

How to control blossom-end rot

Blossom-end rot can be minimized but is hard to totally eliminate.

- Keep the soil moist through regular watering.
- Apply mulch to retain moisture between watering.
- Apply fertilizer according to the results of a [soil test](#).
- Avoid injuring roots; don't cultivate within 1 foot of the base of the plant.
- If you notice that a variety is particularly affected, choose a different variety next year

- Remove fruit with blossom end rot immediately. They will not develop normally, and removing them allows the plant to put its energy into healthy fruit.

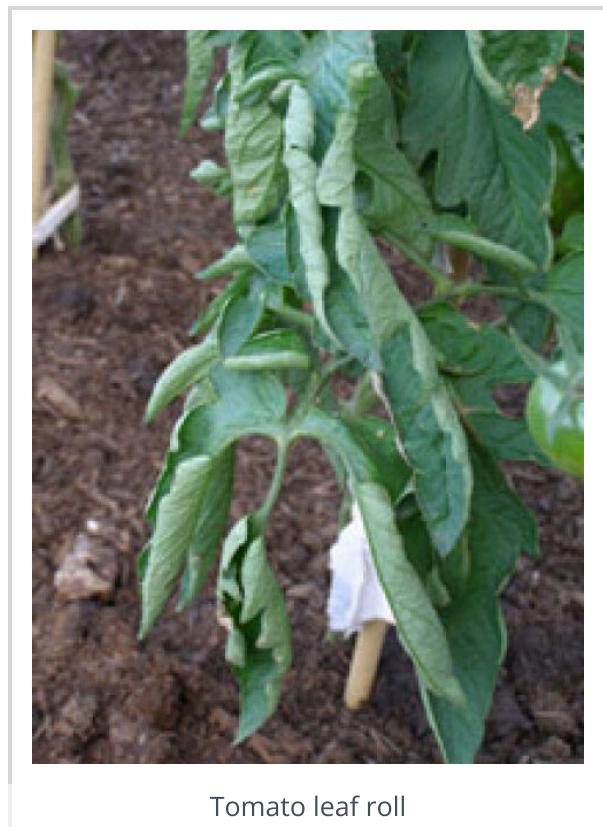
Leaf roll

Leaf roll is a physical disorder of tomatoes associated with hot dry weather but it can occur in response to other stresses like fast growth and pruning. This disorder is believed to be a strategy to conserve moisture.



How to tell leaf roll from other tomato issues

- Leaf margins roll upward until they touch or overlap.
- Affected leaves are firm and leathery to the touch.
- Lower leaves are commonly affected first.
- Once leaves roll, they will not unroll even if weather conditions become cool and wet.
- In severe conditions, the entire plant may exhibit leaf roll.
- Leaf roll does not noticeably reduce plant growth or yield.
- Some varieties exhibit leaf roll more easily than others.
- Leaf roll is very common in tomatoes grown in hoop houses.



How to manage leaf roll

Leaf roll does not reduce plant growth or tomato yield, so no management is necessary.

Sunscald

Sunscald occurs on tomato fruit that have been exposed to too much sun. This is common in plants that have lost leaves from a leaf spot disease or insect feeding, but can also occur on plants that are over pruned or on fruit that are otherwise exposed to the sun.



How to tell sunscald apart other tomato issues

- Sunscald results in a pale yellow to white spot on the side of the fruit facing the sun.
- This area may become a flattened, grayish-white spot.
- The surface may dry out to a paper-like texture.
- Sunscald spots are frequently invaded by decay-causing fungi and bacteria that further rot the fruit.



Sunscald infected by opportunistic pathogens

How to manage sunscald

The best way to avoid sunscald is to maintain a healthy tomato plant by managing insects and diseases that destroy or eat tomato leaves.

Fruit with sunscald will not recover, so remove them to prevent diseases from invading the affected fruit.

Growth cracks

Extremely fast fruit growth can cause growth cracks. This may be caused by periods of abundant rain and high temperatures or can happen when it rains or you water plants after a period of drought.



How to tell growth cracks from other tomato issues

- Cracks may radiate from the stem end of the fruit or may encircle the fruit.
- Cracks are often invaded by secondary fungi and bacteria that further rot the fruit.



Tomato growth crack

How to manage growth cracks

- Maintaining even moisture by watering regularly and mulching the soil around the tomato plant can help reduce growth cracks.
- Varieties differ in susceptibility to cracking, and variety descriptions may be helpful in choosing a plant less likely to crack.

Catfacing

Catface is a condition involving malformation and scarring of fruits, particularly at the blossom end.



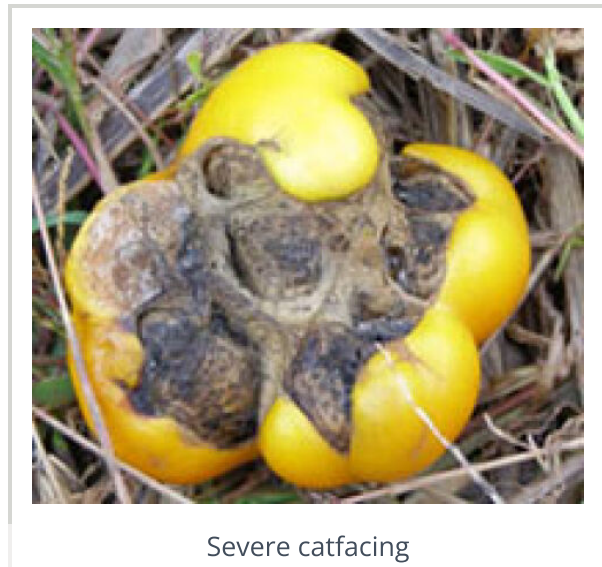
How to tell catfacing from other tomato issues

- Affected fruit are often somewhat flat with a corky brown scar covering the base of the fruit.
- Catfaced fruit can have cavities extending deep into the flesh.

Biology

The causes of catfacing are not definitely known, but it may be caused by:

- Any disturbance to flowers or flower buds.
- Cold temperatures.
- Contact with hormone-type herbicide sprays.



Severe catfacing

How to manage catfacing

- Large tomatoes are more susceptible to catface than small tomatoes.
- Some varieties are particularly prone to catface and should be avoided if it has been a problem in the past.
- Fruit with mild catfacing symptoms are still safe to eat if the fruit has not been infected with secondary pathogens.

Yellow shoulders

Yellow shoulders refer to when the top area on tomato fruit ("shoulder") never ripens, staying hard and yellow or green even as the rest of the fruit is red. Tomatoes can also have issues ripening on the inside, with the inside flesh being white and hard. No matter how long these tomatoes are left on the vine, the shoulder and interior do not ripen.



Biology

Scientists don't fully understand what triggers yellow shoulders.

Yellow shoulders are often seen during and after periods of hot weather (90° F). Other suspected causes involve the interaction between plant nutrition, variety, and plant viruses.

How to manage yellow shoulders

- Do a [soil test](#) and make sure your tomatoes are getting adequate nutrition.
- If you notice a variety with a lot of yellow shoulder, try a different variety next season.

Herbicide injury

The most common injury symptoms are caused by phenoxy herbicides such as 2,4-D (2,4-dichlorophenoxyacetic acid) and dicamba (substituted benzoic acid). These chemicals are growth regulators; they mimic plant hormones.

Tomato plants usually come in contact with the chemical through spray drift or the use of a sprayer that was previously used to apply an herbicide. It is also possible to expose tomato plants to broadleaf herbicides by using grass clippings from lawns recently treated for these weeds as mulch in the vegetable garden.

How to tell herbicide injury from other tomato issues

Affected plants show one or more of the following symptoms depending on the degree of exposure and age of the plant at exposure.

- Older leaves are excessively shriveled, pointed, down-curved, or rolled with prominent light-colored veins.
- Young leaves do not fully expand and are narrow and elongated with parallel veins.



- Distinct yellowing on new leaves where they meet the stem.
- Stems are split, distorted, or brittle.
- Fruits are catfaced or irregularly shaped.
- Symptoms can look very similar to those caused by viruses.
 - Distribution issues within the garden and conversations with neighbors can help determine if viruses or herbicides are causing plant distortion.
- Usually if an herbicide is the cause of damage, other broadleaf plants in different families will show symptoms as well.

Glyphosate injury in tomato foliage



Tomato fruit with herbicide injury

How to manage herbicide injury

- Plants exposed to small amounts of herbicides will outgrow the symptoms without seriously reducing yield or fruit quality. Harvest may be delayed.
- Plants do not recover from severe damage by herbicides.
- If using lawn herbicides, follow label instructions carefully, especially in regards to weather.
- Be sure to follow all herbicide label directions regarding the use of treated grass clippings for mulches in vegetable gardens.
 - While there is usually little threat of injury once the lawn has been mowed four to six times after the herbicide was applied if you are still concerned, leave the clippings on the lawn where they can decompose and provide some nutrients and organic matter back to the lawn.
- Farmers should talk to their neighbors and make sure that anyone who might spray nearby knows that they are growing sensitive crops.
 - [DriftWatch](#) is a tool that allows you to list your farm as a sensitive site; pesticide applicators are supposed to check the site before they spray.

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